

Project AMI – Status Report #2

Team TheParents

CEN4090L – Software Engineering
Capstone

Florida State University

Team Members

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- Nicolas Fernandez (NGF22)

Project Description

- Our team (called TheParents) plan to create an interactive AI model that focuses on the topics in personalized medicine.
- • The AI will be called AMI (Artificial Medical Intelligence).
- • Our goal is to create a functional LLM that provides source transparency.
- • The AI will respond to a given prompt and provide the list of sources that were used in the response. It will provide a citation and potential link to the source.

Assignment of Responsibilities of the Week

- Stefano Buglione:
 - Dataset and source evaluation
- Lauryn Brown:
 - AI model tool feasibility
- Marie Schwarz:
 - Bias and Hallucination Risk Analysis
 - Finalize system requirements
- Nicolas Fernandez:
 - Analysis of AI medical systems

Analysis of Existing AI Medical Systems

- Reviewed MedGemma, BioMistral-7B, LLaVA-Med
 - Compared SLMs vs LLMs performance
 - SLMs ~77% accuracy vs LLMs 88–91% benchmark
 - Smaller models more prone to hallucination
 - AMI will narrow scope to personalized medicine

of existing AI systems

Abstract

Artificial intelligence (AI) has impacted the medical field with a variety of tools and virtual assistants (VAs). As AMI is under development, there must be an analysis of how AI was implemented in the medical field, how language models perform, large language models, and what are the issues that can come up with AMI analysis of open-source language models (SLMs), there can be further considerations before designing AMI. According to some sources, locally run SLMs run poorly compared to language models (LLMs) such as Gemini because of the large gap of computational power. Why the design and implementation of AMI needs to be carefully considered for the use of personalized medicine to avoid hallucinations. The review of SLMs shows that AMI can use for the design and implementation of the architecture and workflow.

Keywords: AI (Artificial Intelligence), ML (Machine Learning), AMI (Artificial Medical Intelligence), LLM (Large Language Model), LMM (Large Multimodal Model), open-Source Language Model, LM (Language Model), VA (Virtual Assistant)

Bias Risks and Hallucinations

- Dataset imbalance (rare diseases underrepresented)
 - Western population overrepresentation
 - Conversational datasets lacking citations
 - Interface over-trust due to friendly avatar design

1. Overview

Artificial Medical Intelligence (A.M.I.) is a conversational AI assistant designed to provide educational information on personalized medicine while maintaining transparency through peer-reviewed citations and ethical safeguards. Because A.M.I. relies on large language models (LLMs) and multiple medical datasets, there are inherent risks of bias, hallucinations, misinformation, and unsafe outputs. Hallucinations occur when an AI generates responses that appear correct but are unsupported or incorrect, while bias occurs when the AI reflects imbalances or limitations in its training data.

2. Bias Risks

Dataset Bias

MedQuAD focuses primarily on general consumer health information, which may overrepresent common conditions and underrepresent rare diseases.

PubMedQA uses peer-reviewed abstracts, but abstract-level information may lack full clinical detail.

HealthCareMagic-100k and MedDialog contain conversational doctor-patient responses without verified citations, which may introduce subjective or outdated information.

MedAESQA provides peer-reviewed citations but has limited size, reducing coverage.

Population Bias

Dataset and Source Evaluation

Dataset and Source Evaluation

1. MedQuAD dataset:
 - Main source: National Institutes of Health (NIH)
 - Has more than 40,000 QA elements
 - HIPAA compliant (has no patient information)
 - Information comes directly from NIH, not from peer-reviewed research journals
 - Focuses on general consumer health topics
2. PubMedQA:
 - Its based on PubMed article Abstracts
 - Answer mainly yes/no/maybe questions
 - HIPAA compliant
 - All sources are peer-reviewed
3. HealthCareMagic-100k:
 - Contains patient-doctor QA
 - Is in conversational format
 - HIPAA Compliant
 - The content is not peer-reviewed, but user generated
 - Does not contain sources to cite.
4. MedDialog:
 - Contains large-scale doctor-patient conversations
 - Not originally in English
 - Good conversational flow
 - HIPAA complaint (conversations are de-identified, but we should still check just in case)
 - Does not contain sources to cite, only conversations
5. MedAESQA:
 - Contains medical questions with manually annotated answers
 - HIPAA compliant
 - Every answer contains a citation to a peer-reviewed article
 - Mostly public health queries
 - Relatively small dataset

Note: for better performance and minimize hallucinations we should also use RAG over a collection peer-reviewed articles like (PMC FTP bulk download).

Avatar Tool Comparison Summary

Feasibility Analysis – AI Avatar Tools Comparison

Tool / Platform	Type	Web Integration	Development Complexity	Performance Impact	Cost	Feasibility for AMI
Blender (Custom 3D Model)	3D Modeling & Rigging	Difficult (requires WebGL/Three.js export)	High	High	Free software	Low–Medium
Live2D	2D Rigged VTuber Model	Moderate (SDK needed for web use)	Medium–High	Medium	Paid license for advanced use	Medium
VTuber Studio / External VTuber Software	Real-time Avatar Software	Not web-native (requires streaming setup)	High	Medium–High	Paid / platform dependent	Low
Lightweight Web Animated Avatar (SVG/Canvas/JS)	2D Web Animation	Very Easy (native browser support)	Low	Low	Free	Very High

Updated Timeline

Task #	Activity Title	Tasks	Assigned To	Start Date	Due Date	Status	Completion %
1	Team & Project Creation	Team Recruitment	-	01/05/26	01/07/26	Complete	100%
		Project Idea	-	01/05/26	01/12/26	Complete	100%
		Project Prospectus	-	01/12/26	01/23/26	Complete	100%
		Group Meetings	-	Saturday	8PM EST	In Progress	47%
2	Requirements Phase	Define system scope and constraints	-	1/12/2026	1/30/2026	Complete	100%
		Identify user roles and use cases	Nicolas Fernandez	2/1/2026	2/8/2026	Complete	100%
		Formalize functional requirements	Stefano Buglione	1/30/2026	2/5/2026	Complete	100%
		Formalize non-functional requirements(LLM, ethics, and privacy laws)	Lauryn Brown	2/4/2026	2/9/2026	Complete	100%
		Draft initial SRS (Software Requirements Specification)	Marie Schwarz	2/6/2026	2/14/2026	Complete	100%
		Define AI avatar requirements	Stefano Buglione	2/6/2026	2/14/2026	Complete	100%
3	Analysis Phase	Analyze existing AI medical systems	Nicolas Fernandez	2/15/2026	2/20/2026	In Progress	100%
		Dataset and source evaluation	Stefano Buglione	2/15/2026	2/25/2026	In Progress	100%
		Analyze bias and hallucination risks	Marie Schwarz	2/18/2026	2/28/2026	In Progress	100%
		Feasibility for avatar tools analysis	Lauryn Brown	2/20/2026	3/1/2026	In Progress	100%
		Finalize system requirements	Marie Schwarz	2/20/2026	3/5/2026	In Progress	50%
4	Design Phase	System architecture design	-	3/6/2026	3/12/2026	Not Started	0%
		LLM training pipeline and deployment design	-	3/6/2026	3/18/2026	Not Started	0%
		UI/UX wireframes	Nicolas Fernandez	3/10/2026	3/20/2026	Not Started	0%
		Design AMI avatar appearance and behavior	Lauryn Brown	3/6/2026	3/22/2026	Not Started	0%
		Design source transparency and citation system	Stefano Buglione	3/15/2026	3/25/2026	Not Started	0%
5	Implementation Phase	Set up chatbot framework	-	3/21/2026	3/28/2026	Not Started	0%
		Implement response generation	-	3/25/2026	4/2/2026	Not Started	0%
		Integrate APIs	-	3/28/2026	4/5/2026	Not Started	0%
		Implement text-to-speech	-	3/30/2026	4/3/2026	Not Started	0%
		Frontend implementation	-	4/1/2026	4/5/2026	Not Started	0%
6	Test Phase	Acquiring additional test data (using PRISMA)	-	3/27/2026	4/3/2026	Not Started	0%
		Drafting test cases	-	3/27/2026	4/2/2026	Not Started	0%
		Running test cases	-	4/1/2026	4/7/2026	Not Started	0%
		Evaluate chatbot performance (accuracy, latency)	-	4/3/2026	4/9/2026	Not Started	0%
		Conduct internal usability testing and team review	-	4/5/2026	4/10/2026	Not Started	0%
7	Delivery	Due Date	-	4/17/2026	4/17/2026	Not Started	0%
		Project Demonstration	-	4/10/2026	4/17/2026	Not Started	0%
		Final SRS	-	4/3/2026	4/10/2026	Not Started	0%
		Peer Review	-	4/10/2026	4/17/2026	Not Started	0%

Planned Activities

- Finalize system requirements
 - Begin design phase with LLM training(TBD), UI/UX wireframes(Nicolas Fernandez), designing the avatar and behavior(Lauryn Brown), source transparency and citation(Stefano Buglione), and system architecture(TBD).

Issues Not Resolved

- Final dataset composition decision
 - Accuracy validation method (90% target)
 - Avatar implementation tool selection
 - Stretch goal: user accounts functionality

Summary

- Requirements nearing completion
 - Analysis documentation completed
 - Risk mitigation plan established
 - Transitioning into Design Phase